

Event- and annual-scale precipitation extremes enhance groundwater recharge at the ecological restoration catchment of hilly and gully region

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ABSTRACT

Precipitation extremes and ecological restoration projects significantly influenced hydrological processes by mitigating or aggravating groundwater depletion within the Earth's critical zone. However, scientific evidence remains limited due to the strong dependence of deep recharge on both unsaturated zone thickness and precipitation event magnitude. Here, we analyzed seven-year field datasets encompassing precipitation, soil water content (SWC), surface water (reservoir water), and groundwater from an ecological restoration catchment on the Chinese Loess Plateau. Precipitation extremes triggered deep hydraulic connectivity between the unsaturated and saturated zones mainly through preferential flow pathways, as evidenced by depleted $\delta^{18}\text{O}$, increased of SWC profiles, and rising water table. Temporal-spatial patterns of SWC (0–4000 cm profile) revealed preferential flow pathways on a sunny slope with recharge efficiency regulated by precipitation patterns and topography. In the gully, water tables showed a positive correlation with precipitation amount, duration, and initial SWC. These hydrological drivers induced significant differences in water table changes among precipitation years. In contrast, the slope (0–500 cm profile) maintained persistent water deficits with limited recharge response, despite precipitation inputs (except for a 167.7-mm event). Furthermore, precipitation variability coupled with plant root uptake altered the vertical soil water gradient, with the 200–300 cm layer functioning as a hydraulic buffer. However, prolonged drought triggered an accelerated water table recession at 60.8 mm/year, while creating carry-over soil water deficits that extended beyond dry years, sustaining 15.4 mm/year depletion even in the following normal year. When considering the saturated zone contribution, total recharge rates demonstrated the positive hydrological feedback of ecological restoration projects to annual precipitation, accounting for 45 % of precipitation (327.8 mm/year) during a wet year. Therefore, it can be inferred that event- and annual-scale precipitation extremes enhance groundwater recharge at the ecological restoration catchment. These findings provide critical scientific support for maintaining groundwater sustainability in such systems.

1. Introduction

Groundwater, the most plentiful source of freshwater on Earth, plays a crucial role in feeding both livelihoods and healthy ecosystem developments within Earth's critical zone (between the canopy and the base of groundwater) (Saccò et al., 2024; Singha et al., 2024). Nevertheless, the impacts of climate change and human activities pose a formidable threat to global groundwater resources, including drought, excessive withdrawal and contamination in the Anthropocene (Bierkens

and Wada, 2019; Hartmann et al., 2021; Zhu and Zhang, 2022). Of particular concern is the widespread water table decline and aquifer exhaustion resulting from an imbalance between recharge and discharge (Jasechko et al., 2024). When the recharge is insufficient or groundwater storage is depleted over an extended period, groundwater resources become unsustainable (Famiglietti, 2014). Therefore, understanding groundwater recharge is essential for ensuring a sustainable supply of renewable groundwater. Nonetheless, accurately assessing groundwater recharge is challenging due to its dynamic

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nature, compounded by a general scarcity of measurement data (Moeck et al., 2020).

Recharge refers to the diffuse or localized inflow of water into an aquifer system (Alley et al., 2002). Due to its inherently unstable nature and sensitivity to a variety of climatic and non-climatic factors, recharge exhibits high heterogeneity in both time and space, ranging from 9 mm/year to 238 mm/year globally (Barron et al., 2012; Fu et al., 2019; Moeck et al., 2020; Zhang et al., 2023). Spatially, recharge is particularly sensitive to climate change in semi-arid regions, where even a modest increase in the aridity index can significantly reduce recharge (Turkeltaub and Bel, 2023; Berghuijs et al., 2024). Furthermore, strong temporal variability is observed, with recharge during extremely wet years being 6.5 times larger than during extremely dry years, primarily driven by event-scale precipitation extremes (Li et al., 2018a; Gao et al., 2023; Hu et al., 2023). In general, groundwater flux requires about a century to achieve a new equilibrium in response to climate-induced recharge variations (Cuthbert et al., 2019). However, rapid water table fluctuation has been observed in semi-arid regions with shallow groundwater systems, exhibiting a 35-day lagged response to extreme precipitation events (Zhao and Wang, 2021). These divergent response times are common in the hilly and gully regions where groundwater is becoming increasingly important for both human use and ecosystems (Tan et al., 2017). Therefore, it is crucial to determine when groundwater recharge occurs and how much precipitation is required to trigger it, particularly in areas affected by human activities (Priestley et al., 2025).

There is growing evidence that human activities are intensifying the water cycle through alterations in the structure and characteristics of terrestrial planetary surfaces, as well as through land use practices such as ecological restoration (Abbott et al., 2019; Kuang et al., 2024). Compared with the feedback time of groundwater to climate change, the response time of human-induced ecological restoration is orders of magnitude shorter (Cuthbert et al., 2019; Zhao et al., 2019). In the Global North, afforestation has been increasing over the past several decades. This expansion has influenced land use change across 32 % of the global land area (Peng et al., 2014; Winkler et al., 2021). In China, the most significant land use changes have been observed on the Loess Plateau (LP), primarily attributed to large-scale ecological restoration such as the Grain for Green (GFG) and the Gully Land Consolidation (GLC) projects (Chen et al., 2020; Wang et al., 2024). These ecological restoration projects lead to profound shifts and dynamics in hydrological processes (Li et al., 2018b; Hou et al., 2023). Notably, the GFG project enhanced evaporation and precipitation recycling processes, and amplified soil water deficits over time, even with reduced recharge (Sun et al., 2024a; Wang et al., 2024). Particularly in water-limited regions, terrestrial water and groundwater storage depleted at the rates of 4.9 mm/year and 7.6 mm/year, respectively (Chen et al., 2025). It's worth noting that abrupt changes of these hydrological effects occurred around 2010 (Zhu et al., 2025). Concurrently implemented in 2011, the GLC project demonstrated distinctive hydrological responses, characterized by a rising water table, sustained soil water storage within the upper 300 cm profile, and well-defined vertical soil water gradients (Jin et al., 2019; Zhao et al., 2022; Ma et al., 2025). These divergent hydrological responses to ecological restoration are driven by high spatial and temporal heterogeneity of recharge, the uncertainty of potential recharge estimates, and its ongoing instability following human disturbance. Therefore, evaluating actual recharge through direct field measurement is a key challenge, especially in thick unsaturated zones.

On the LP, recharge must first traverse the variably thick overlying unsaturated zone, where spatio-temporal heterogeneity of hydrological processes and low water fluxes predominantly control recharge dynamics. These mechanisms are further modulated by precipitation patterns and ecological restoration efforts. Meanwhile, the dominant recharge mechanism controlled by unsaturated zone thickness varies with topography: piston flow in tablelands versus preferential flow dominant in gullies (Xiang et al., 2019). Extensive literature delineates

the potential recharge ranging from decades to hundreds of years on the LP (Huang et al., 2020; Wang et al., 2023b). This protracted timescale of recharge is attributed to the loess structure's low hydraulic conductivity, variations in the thickness of unsaturated zone and depth-limited development of preferential flow paths (< 4000 cm) (Li et al., 2019; Huang et al., 2022; Wang et al., 2023b). Notably, in the hilly and gully regions, field monitoring depth can broaden to water table, and groundwater responds readily to sub-annual and annual precipitation (Tan et al., 2017). Therefore, this study focuses on a representative loess catchment in the hilly and gully region to quantify the impacts of precipitation and ecological restoration on groundwater recharge. We aim to address the following questions: (1) Is there hydraulic connectivity in the hilly and gully regions with extensive ecological restoration projects? (2) What is the characteristic response of water table to event- and annual-scale precipitation extremes? (3) What hydrological consequences have emerged from precipitation extremes and ecological restoration projects?

2. Materials and methods

2.1. Study area

The field experiment was conducted in the Gutun watershed (36°46'45"N, 109°48'20"E), Yan'an City, Shaanxi Province, China, near the middle reaches of the Yellow River (Fig. 1). The watershed covers approximately 25 km², where the ecological restoration such as the GFG (since 1999) and the GLC (since 2011) projects has been implemented. The study area is characterized by a monsoon climate, with average annual values from 2017 to 2023 of 555 mm in precipitation (occurring mostly from June to October, i.e., the rainy season), 11 °C in temperature, and 868 mm/year in evapotranspiration rate (Fig. 2). The ratio of potential evapotranspiration to precipitation indicates that this region is water-limited. Influenced by ecological restoration efforts, micro-topography has been reconstructed, resulting in land use changes such as the establishment of forest and grass on the slopes, as well as farmland in the gullies. These changes have intensified the spatial and temporal patterns of soil water and created new hydrological conditions (Wang et al., 2019). Due to variations in loess thickness and elevation (ranging from 973 m to 1188 m), the groundwater depth shows a significant difference between the slopes (> 40 m depth) and the gullies (< 10 m depth). Therefore, groundwater is largely inaccessible to vegetation on the slopes. Additionally, a reservoir (representing surface water) at the gully outlet following the GLC project was created approximately 200 m horizontally from the nearby monitoring well 4. The reservoir's water table remains consistently lower than the water table at well 4, which makes groundwater recharge from this surface water source unlikely.

2.2. Experimental design and sample collection

In this study, field monitoring of hydrological systems for seven consecutive years was conducted at the Shaanxi Loess Plateau Earth Critical Zone National Field Observation and Research Station. The hydrological systems include meteorological station, surface water, soil water (neutron probe, CNC100, Probe Technology Co., Ltd., China), and groundwater monitoring (HOBO sensor, U20-001-04, Onset Computer Corp., USA) (Fig. 1). The liquid water samples included precipitation from each precipitation event (from 2018 to 2022, $n = 73$), soil water from 3000 cm profiles (collected at 20 cm, 40 cm, 100 cm, 200 cm, ..., 3000 cm layers during 2021, $n = 64$), surface water (2018 and 2019, $n = 12$), and groundwater (2018 and 2019, $n = 12$). It was noteworthy that the soil water was first extracted using an automatic cryogenic vacuum distillation water extraction system (LI-2100, LICA Inc., Beijing, China). All of the liquid water samples were filtered (0.22 μm), and then the values of $\delta^2\text{H}$ and $\delta^{18}\text{O}$ were determined using Los Gatos Research liquid-water isotope analyzer (LGR Inc., California, USA) at the Institute of Earth Environment, Chinese Academy of Sciences. The results are

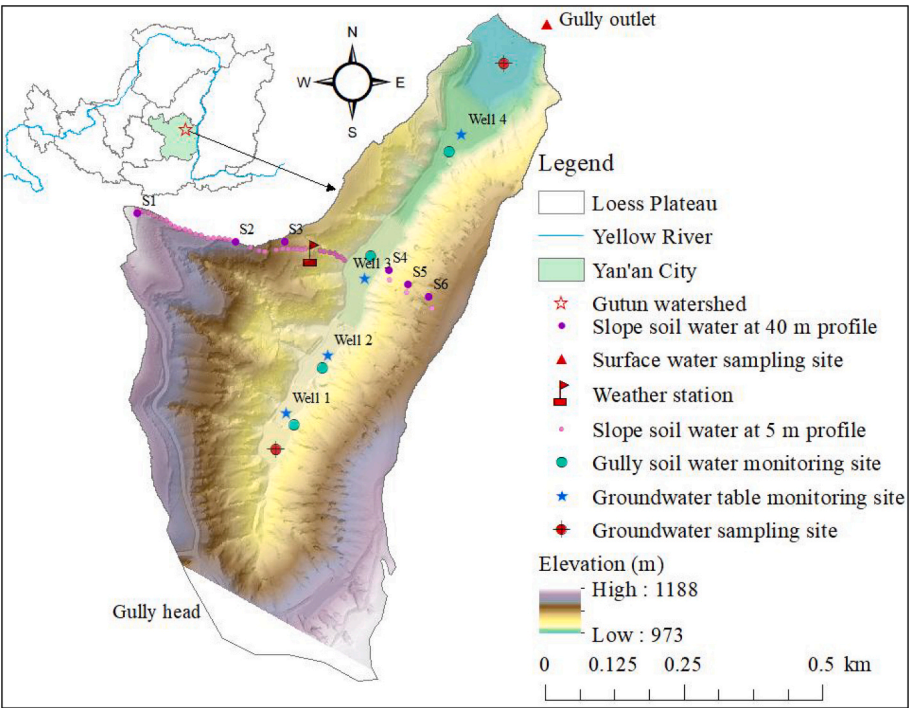


Fig. 1. Location of study area and sampling sites.

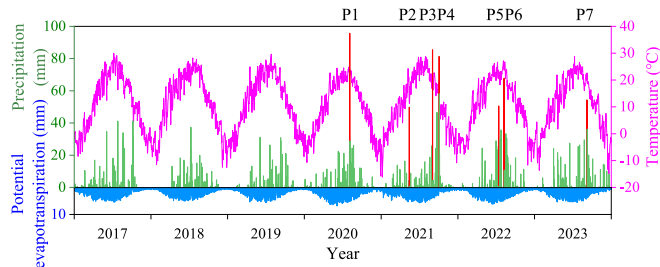


Fig. 2. Time series of precipitation (P represents precipitation event), potential evapotranspiration and temperature in the study area from 2017 to 2023.

reported as deviations from the Vienna Standard Mean Ocean Water (VSMOW) in per mil differences using delta notation (precision is ± 0.1 ‰). Volumetric SWC was measured monthly by installing aluminum neutron probes (500 cm ($n = 42$) or 4000 cm ($n = 6$)) at 20-cm intervals from 2019 to 2022. Distributed soil samples were collected from 0 to 4000 cm profiles at 50 cm intervals and analyzed using a particle size analyzer (Mastersizer 3000, Malvern, England, UK). The contents of clay (< 0.002 mm), silt (0.002–0.05 mm), and sand (> 0.05 mm) were calculated as percentages (Table 1). Five HOBO sensors were installed in the gully from the head to the outlet to record air ($n = 1$) and groundwater ($n = 4$) pressures and to calculate groundwater table (well 1, well

2, well 3, and well 4) from 2017 to 2023.

2.3. Data analysis

2.3.1. General description of recharge, precipitation years, and event- and annual-scale precipitation extremes

In alignment with the definition of groundwater recharge and the soil water balance principles, this study defines recharge as the vertical percolation of soil water through profiles to replenish both the unsaturated and saturated zones. Specifically, recharge processes are triggered when soil water storage (SWS) increases and/or the water table rises. Based on the precipitation threshold of deep recharge ($>$ annual infiltrated depth) (Xiang et al., 2020), precipitation extremes at the event-scale are defined as daily precipitation exceeding 30 mm/day and cumulative precipitation surpassing 50 mm. Following this criterion, seven precipitation events (P1–P7) were identified using climate data from the local weather station from 2017 to 2023, with detailed information provided in Table 2 and Fig. 2. For this study, precipitation year was classified according to deviations of ± 25 % from the long-term annual average precipitation (555 mm). Wet, normal, and dry years corresponded to annual precipitation values of ≥ 694 mm, 426–694 mm, and ≤ 426 mm. During the study period, two dry (2018: 405 mm, 2019: 406 mm), four normal (2017: 557 mm, 2020: 556 mm, 2022: 647 mm, and 2023: 583 mm), and one wet (2021: 731 mm) years were recorded, respectively. In addition, in this study, recharge soil layer was defined as

Table 1
Basic soil physical properties, topography parameters, sampling depth, and land use type.

Sampling position			Soil properties			Topography parameters			Sampling depth (m)	Land use type
			Clay (%)	Silt (%)	Sand (%)	Slope (°)	Aspect (°)	Elevation (m)		
Sunny slope	Top	S1	8.5	74.5	17.0	18	93	1176	40	Forest
	Middle	S2	10.5	72.2	17.3	15	134	1138	40	Shrub
	Down	S3	10.9	75.2	13.9	22	76	1080	40	Forest
Shade slope	Down	S4	9.2	75.5	15.3	37	339	1009	10	Forest
	Middle	S5	8.8	74.0	17.2	27	315	1049	33	Forest
	Top	S6	8.8	72.2	17.0	20	253	1071	34	Grass

Table 2

Detail information of the seven extreme precipitation events.

Precipitation event	Time	Total amount (mm)	Peak intensity (mm/h)	Duration (day)
P1	5–6 August 2020	124.6	28.6	2
P2	15 May 2021	49.8	5.8	1
P3	3 September 2021	85.5	23.1	1
P4	4–5 October 2021	139.6	7.5	2
P5	14–15 July 2022	81.8	18.6	2
P6	9 August 2022	67.8	60.4	1
P7	8–9 September 2023	90.6	22.6	2

the soil layer where the change of soil water storage exceeds the average across the profile.

2.3.2. Potential evapotranspiration

To accurately calculate the actual recharge of groundwater, potential evapotranspiration (PET, mm/d) must be considered using ET₀ calculator software (version 3.2) based on the FAO Penman-Monteith equation (Eq. 1).

$$PET = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T+273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \quad (1)$$

where R_n , G , γ , T , u_2 , e_s , e_a , and Δ represents net radiation at the crop surface (MJ/m² day), soil heat flux density (MJ/m² day), psychrometric constant (kPa/°C), mean daily air temperature at 2 m height (°C), wind speed at 2 m height (m/s), saturation vapor pressure (kPa), actual vapor pressure (kPa), and the slope of the vapor pressure curve (kPa/°C), respectively.

2.3.3. Soil water storage

The SWS (mm) per unit area and its change (Δ SWS, mm/year) were described by eq. 2 and 3, respectively:

$$SWS = \sum_{i=1}^n \theta_i * d_i * 10 \quad (2)$$

$$\Delta SWS = SWS_D - SWS_J \quad (3)$$

where θ_i , d_i , n , SWS_D , SWS_J , and 10 represents the average SWC at depth i (cm³/cm³), the thickness of the soil layer i (cm), the total number of soil layers, SWS in the end (December) and the beginning (January) of the year (mm), and unit conversion factor from cm to mm, respectively.

2.3.4. Water-table fluctuation method

Groundwater recharge (R, mm/year) was calculated by using water-level fluctuation method (Healy and Cook, 2002) (Eq. 4):

$$R = S_y * \frac{\Delta h}{\Delta t} * 10 \quad (4)$$

where S_y , Δh , Δt , and 10 represents the specific yield, change of water-table height (cm), time (year), and unit conversion factor from cm to mm, respectively. Theoretically, the S_y can be calculated using eq. 5. However, in this study area, particularly within the gully, the initial SWC above the water table usually approaches saturation. Under such condition, even minor water input may trigger a rapid rise in the water table, resulting in an overestimation of recharge. Therefore, a corrected S_y^* value is recommended for incorporation into eq. 6 (Gong et al., 2023).

$$S_y = \frac{V_{\text{free}}}{V_{\text{soil}}} = \frac{V_{\text{total}} - V_{\text{bound}}}{V_{\text{soil}}} = \theta_s - \theta_r \quad (5)$$

$$S_y^* = \theta_s - \theta_l \quad (6)$$

where V_{total} , V_{free} , V_{bound} , V_{soil} , θ_s , θ_r , and θ_l represents the volume of total water, free water, bound water, and soil volume (cm³), saturated, residual, and initial SWC (cm³/cm³), respectively. Notably, to account for spatial heterogeneity and the depth dependence of specific yield, measurements were collected at seven sites along the gully across the 500 cm profiles (at 50-cm intervals) in previous studies (Zhao et al., 2019). The average value of 0.19 derived from these measurements was used to represent the specific yield in the study area, and this value falls within the documented range for the same soil texture type (Lv et al., 2021).

2.3.5. Statistical methods

All data were stored using Excel 2019 (Microsoft Corp., USA). The significant difference test was performed in Origin Pro 2024 (Origin Lab, USA) by using Least Significance Difference (LSD) and Kruskal-Wallis' tests. The map of monitoring sites and other figures were generated using ArcGIS (Arc Map™ 10.2, Esri, USA) and Origin Pro 2024, respectively.

3. Results and Discussion

3.1. Conceptualize hydraulic connectivity in the hilly and gully region

3.1.1. Stable isotopic compositions of different water bodies

Fig. 3 presents the stable isotopic compositions of precipitation, surface water, soil water, and groundwater in this study area. Precipitation isotopes displayed substantial variability, with $\delta^2\text{H}$ ranging from −134.6 ‰ to −3.2 ‰ and $\delta^{18}\text{O}$ ranging from −18.7 ‰ to −0.7 ‰. These ranges align with previously reported values for the LP region (Sun et al., 2020). The temporal variations in precipitation isotopes primarily reflect monsoon precipitation and evaporation effects, as evidenced by significantly negative values recorded during event-scale precipitation extremes in the rainy season (Xiang et al., 2020). The Local Meteoric Water Line (LMWL) derived from 2018 to 2022 precipitation data shows a strong linear relationship: $\delta^2\text{H} = 7.3 * \delta^{18}\text{O} + 5.8$ ($R^2 = 0.96$).

Most surface water, soil water, and groundwater isotopic values plot below the LMWL, suggesting that they originated from local precipitation but have undergone varying degrees of evaporation. Surface water isotopes exhibited a slope parallel to the LMWL, suggesting rapid precipitation recharge with limited evaporation. In contrast, soil water demonstrated significant isotopic enrichment ($\delta^2\text{H}$: −82.3 ‰ to −61.6 ‰; $\delta^{18}\text{O}$: −11.5 ‰ to −8.0 ‰ (excluding the 20 cm depth), reflecting the combined effects of infiltration process, evaporation, new and old water

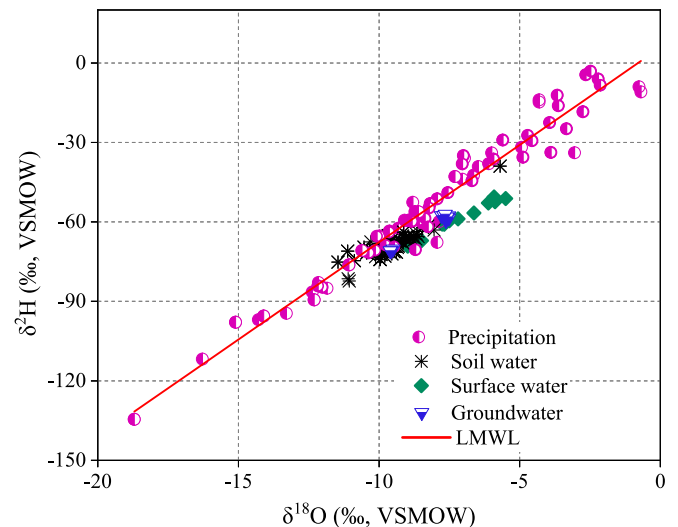


Fig. 3. Dual plots of $\delta^2\text{H}$ and $\delta^{18}\text{O}$ in precipitation, surface water, soil water, and groundwater. LMWL represents the Local Meteoric Water Line.

mixing, and land use impacts (Sun et al., 2024b). Notably, some soil water isotopic values plotted above the LMWL, likely indicating rapid preferential flow prior to evaporation. Similar findings were reported by Huang et al. (2020), who observed more depleted $\delta^{18}\text{O}$ values in soil water compared to annual precipitation. This pattern occurs because the deep soil is recharged by precipitation extremes, whose isotopic values generally resemble those of precipitation during the rainy season. Furthermore, continuous precipitation extremes and preferential flow pathways, such as macropores formed by root growth and decay, enhance the hydrological connectivity between shallow and deep soil layers (Xiang et al., 2019). As a result, preferential flow can recharge soil layers below most of the rooting depth and even reach the groundwater, even though it accounts for only a small fraction of the total recharge (Singha et al., 2024).

Additionally, groundwater exhibits the narrowest isotopic range, despite distinct responses between the head and the outlet of the gully. This spatial pattern suggests faster recharge in the head of the gully areas (Li et al., 2017). The partial isotopic overlap among different water bodies suggests potential hydraulic connectivity in this study area. Integrating horizontal distance and elevation data reveals a direct recharge regime in which groundwater is recharged by both precipitation and soil water, while surface water is recharged by precipitation and groundwater discharge.

3.1.2. Soil water stable isotopic composition distributions in forest and grassland profiles

Fig. 4 shows comparable vertical distributions of $\delta^2\text{H}$ and $\delta^{18}\text{O}$ values in the forest (S3) and grassland (S6) profiles, which share similar elevation, climate conditions, and soil particle composition (Table 1). The isotopic values exhibited medium variability (coefficient of variation, CV: 13–31 %) in the upper 0–100 cm layer and weaker variability (CV: 4–7 %) in the 100–3000 cm layer. Notably, the transition depth at 100 cm corresponded to the annual average infiltration depth of precipitation in the semi-arid ecosystem of the LP. Vertical isotopic gradients of soil water in the profiles were primarily influenced by precipitation infiltration and evaporation processes. During the sampling period (March 2021), cumulative PET (75.3 mm) obviously exceeded precipitation (21.2 mm) (Fig. 2), driving progressive isotopic enrichment and stabilization with depth through the evaporation effect (Xiang et al., 2021).

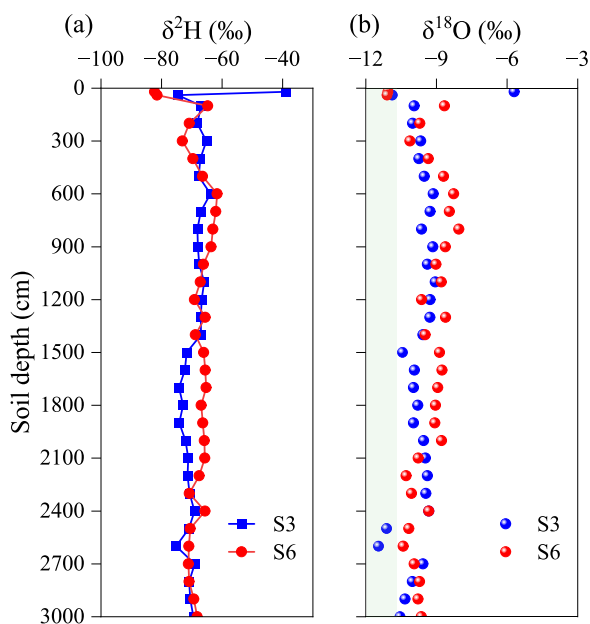


Fig. 4. Soil profiles of soil water stable isotopes in forest (S3) and grassland (S6).

An anomalous depleted $\delta^{18}\text{O}$ peak was observed at 2600 cm depth in the S3 profile. Given that deep soil water in this study area originates solely from local precipitation (Fig. 3), this anomaly suggests preferential flow pathways in the S3 profile. This interpretation was further supported by comparatively depleted isotopic values in the S3 versus S6 profiles. Three lines of evidence substantiate this mechanism: 1) recharge dependent on piston flow was limited to depths above 1000 cm within the thick unsaturated zone even during wet years (Gao et al., 2023; Shi et al., 2021; Tao et al., 2021); 2) the maximum rooting depth of *Robinia pseudoacacia* (2260 cm), historical precipitation tracking (vertical movement since 1963 reaching 2550 cm), and preferential flow capacity (< 4000 cm) collectively enable deep percolation to 2600 cm (Chen et al., 2023; Wang et al., 2023b); 3) despite higher soil water consumption, the deep-rooted vegetation in the S3 profile facilitates faster hydraulic redistribution compared with the shallow-rooted system in the S6 profile, reflecting the effect of root architecture (Li et al., 2024). These findings demonstrate that preferential flow pathways enable precipitation recharge to depth exceeding 2600 cm, as evidenced by isotopic signatures. This process implies an additional water source for deep-rooted vegetation following the GFG project, which could help mitigate severe water deficits in water-limited ecosystems. This notion is supported by a recent study indicating that past precipitation stored in deep soils helps sustain ecological stability in such regions (Shao et al., 2025). Therefore, our insights have significant implications for reducing the risk of drought stress and sustaining eco-hydrological functions.

3.1.3. Temporal and spatial dynamics of soil water content

Fig. 5 demonstrates significant vertical and horizontal spatial heterogeneity in SWC across the study area. The varying thickness of the unsaturated zone explains the complex distribution patterns of average SWC (S1: 11.0 %; S2: 12.9 %; S3: 12.7 %; S4: 13.7 %; S5: 14.6 %; and S6: 19.3 %) in the profiles. The maximum SWC values reached 26.3 % at 3300 cm (S1), 23.5 % at 2950 cm (S2), 26.7 % at 3950 cm (S3), 26.3 % at 960 cm (S4), 24.3 % at 3000 cm (S5), and 30.2 % at 1000 cm depth (S6), showing aspect-dependent differences where sunny slopes exhibited deeper SWC infiltration compared to shady slopes. This spatial pattern supports the preferential flow hypothesis previously indicated by stable isotope analysis in Section 3.1.2. The observed minimum SWC values (5.5–13.3 % across S1–S6 profiles) fell significantly below the stable field capacity in the LP (17.6 %), highlighting critical water deficits in deep soil layers (> 200 cm depth) (Jia et al., 2020). These findings emphasize the importance of optimizing unsaturated zone water storage through combined preferential and piston flow mechanisms to sustain vegetation growth and groundwater recharge (Gao et al., 2024).

Fig. 6 reveals dynamic infiltration patterns where increased precipitation and unsaturated zone SWC enable deep percolation (> 200 cm depth), with wetting front expansion showing temporal dependency. Topographic controls manifest through aspect-driven contrasts, where shady slopes demonstrated higher SWC compared with sunny slopes, attributed to differential solar radiation exposure (Liu et al., 2024). Below the wetting front, a persistent dry gap layer (300–900 cm thickness) reflects precipitation limitation effects, with thickness variations correlating to vegetation type (Wang et al., 2023a). Notably, an SWC of 39.8 % (nearly saturated) was observed at the maximum depth in the S3 profile (September 2021), suggesting preferential flow contribution from 380.5 mm summer precipitation, as evidenced by a vertical discontinuous water channel in Fig. 2 and Fig. 6. This phenomenon's exclusive occurrence on sunny slopes (absent in shady slopes) visually confirms aspect-controlled preferential flow dynamics. In addition, a possible explanation for the maximum SWC (39.7 %) recorded at a depth of 3860 cm (S3 profile) is an elevated water table. This observation contributes to our understanding of the hydrological connectivity between different water bodies.

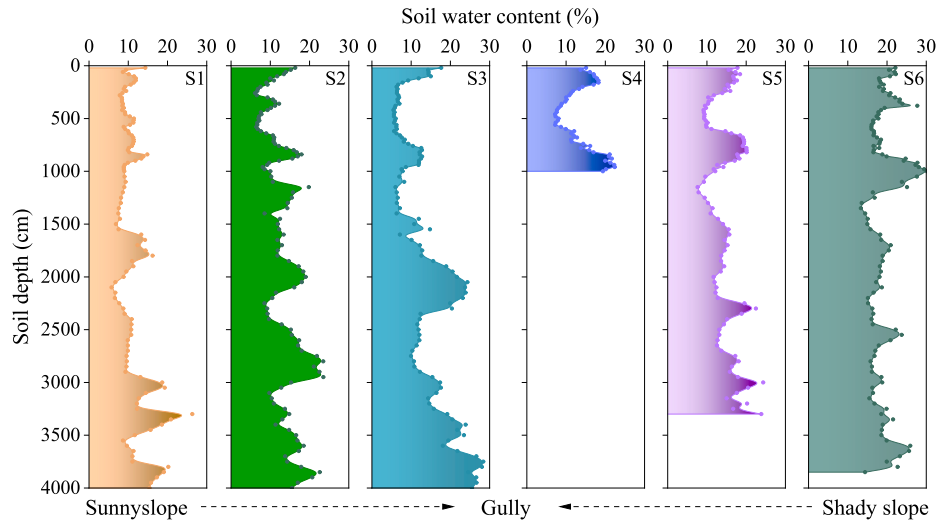


Fig. 5. Vertical distribution of soil water content at the 4000 cm soil profile for both the sunny and shady slopes.

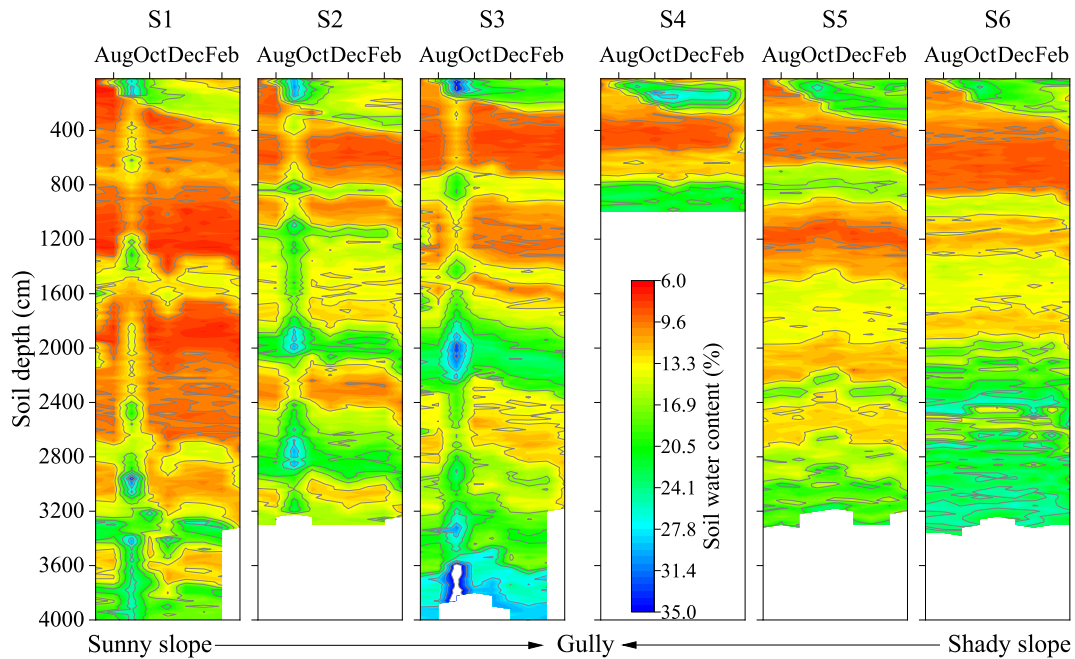


Fig. 6. Dynamics of monthly soil water content at the 4000 cm soil profile from July 2021 to March 2022.

3.2. Response of water table to event- and annual-scale precipitation

3.2.1. Effects of event-scale precipitation extremes

The amplitude of water table fluctuations in wells increased following event-scale precipitation extremes (Fig. 7a). However, these fluctuations in water tables exhibited significant spatial and temporal heterogeneity due to the combined effects of precipitation event characteristics and initial hydrological conditions on recharge efficiency (Stahl et al., 2020). Generally, continuous low-intensity precipitation events demonstrated optimal groundwater recharge potential. For example, measurable recharge responses occurred during P2 and P4 events, with water table rising 6.4–57.7 cm (Table 2 and Fig. 7). In contrast, high-intensity precipitation (> 18.6 mm/h) showed limited recharge efficiency, with water table rising only 0–20.3 cm. Notably, the P3 (85.5 mm, 23.1 mm/h) failed to induce detectable response across wells, likely due to rapid surface runoff generation in low-infiltration soils (Luo et al., 2023b). Therefore, negative correlation was observed

between peak intensity and recharge efficiency, whereas positive associations emerged for precipitation amount and duration (Fig. 7b-d). Interestingly, consecutive dry days and initial water table depth showed no significant correlation with recharge magnitude (Fig. 7e-f), contradicting previous findings emphasizing drought impacts on groundwater systems (Nott et al., 2024). This may be because the variable thickness of unsaturated zone weakens the effects of drought and initial water table conditions on recharge. For example, in shallow water table systems, recharge was typically observed to be positively correlated with initial water table but negatively correlated with consecutive dry days (Jin et al., 2025; Ma et al., 2025). In contrast, recharge in deeper water table systems can be largely unaffected by these hydrological factors over short periods (Sun et al., 2024). Well 4 exhibited minimal fluctuations due to the thick unsaturated zone buffering, where SWS met evaporation demand. Conversely, well 1 showed rapid responses (159.5 cm rise post-P4) with 22-day lag time, suggesting preferential flow dominance over matrix infiltration with a significantly faster response than that reported

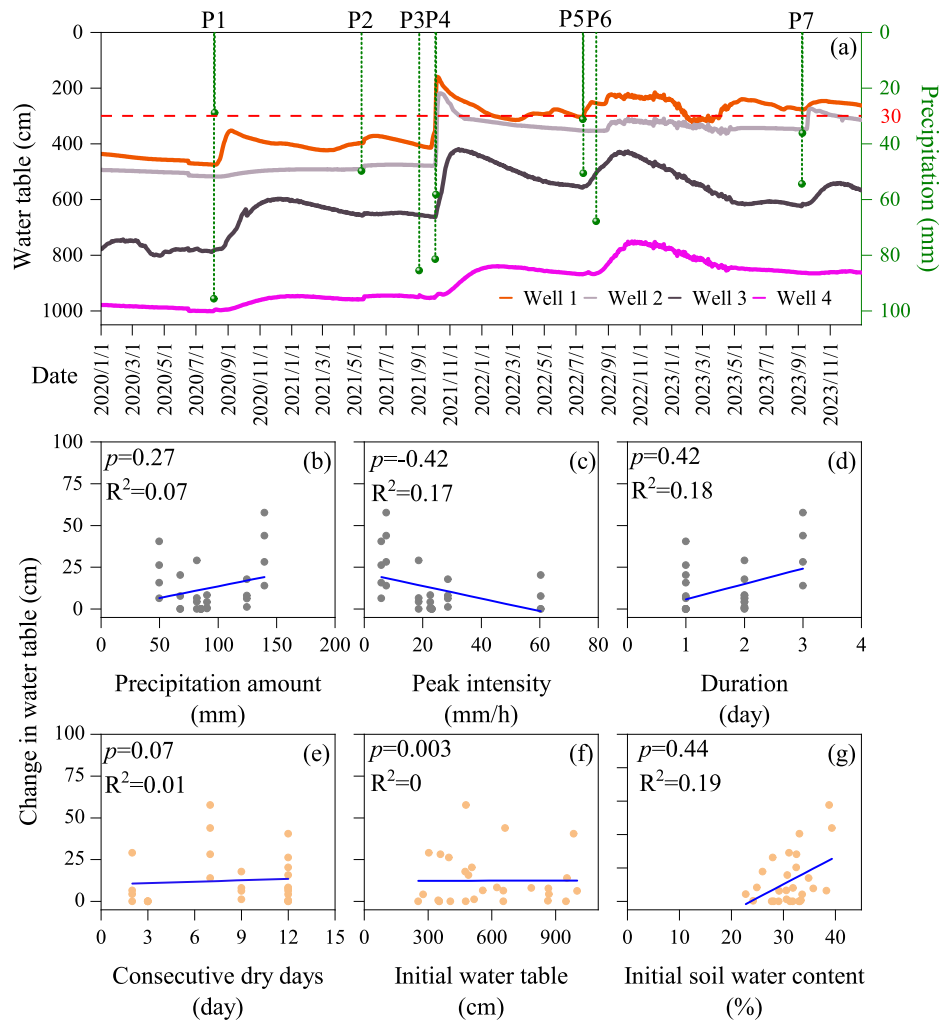


Fig. 7. Time series of water table during precipitation event (a) and relationship between change groundwater table and various factors (b–g). P represents precipitation event.

in the previous study (Zhao et al., 2022). In addition, initial SWC was positively correlated with recharge (Fig. 7g). This may be because the wet conditions induced by the high initial SWC, likely increasing connectivity between draining soil water and groundwater. Therefore, these results not only confirmed that event-scale precipitation extremes could drive groundwater recharge in shallow aquifer systems (Gabrielli et al., 2012), but also indicated preferential flow dominance under climate extremes.

3.2.2. Effects of annual precipitation

The response of water table to annual precipitation exhibited distinct variations across wells, with water table increments measuring 315.4 cm (well 1), 318.9 cm (well 2), 385.3 cm (well 3), and 251.1 cm (well 4) from 2017 to 2023 (Fig. 8). Spatio-temporal analysis revealed a sequential “awakening” pattern along the gully’s topographic gradient, where both response lag times and amplitudes decreased with elevation. These results indicated strong heterogeneity in the response of the water table to precipitation in a very small gully, highlighting the importance of having high temporal and spatial resolution groundwater data to understand the effects of precipitation on recharge (Lorenzo-Lacruz et al., 2017). Three distinct hydrological response patterns emerged corresponding to dry, normal, and wet years (Fig. 8a). During the dry years, recharge was severely limited across all wells except well 3, where additional water from human activities induced fluctuations (red box, Fig. 8a). Maximum water table depths reached 474.6 cm (well 1), 517.1

cm (well 2), 803.7 cm (well 3), and 1001.2 cm (well 4). Notably, in well 1 with shallow water table, vegetation roots continued using deep soil water and groundwater for growth and consequently dropped water table. While water tables generally recovered to their initial levels during normal years, indicating sufficient unsaturated zone storage to sustain local ecosystem. A similar result was reported in the previous study that analyzed soil water data over a five-years period (Zhao et al., 2022). The wet year triggered rapid water table rises of 98.2–176.0 cm, leading to an average increase of water tables by 139.4 cm across the gully. In addition, statistical analysis confirmed significant differences in water tables among dry (2019), normal (2020), and wet (2021) years ($p < 0.05$) (Fig. 8b). Notably, strong synchronization between slope SWC and gully water tables during the rainy season of 2021 revealed direct hydraulic connectivity (Fig. 6 and Fig. 8a). These results demonstrate that precipitation exerts a more immediate and pronounced influence on groundwater recharge dynamics in the hilly and gully regions of the LP than previously recognized (Huang et al., 2018; Berghuijs et al., 2022). Therefore, more hydrological regulation is essential for these systems under both event- and annual-scale precipitation extremes.

3.3. Impacts of ecological restoration on recharge

Recharge efficiency exhibited obvious variability across precipitation events along the slope gradient (Fig. 9). The maximum and minimum recharge efficiencies were observed following events P4 (17.3 %

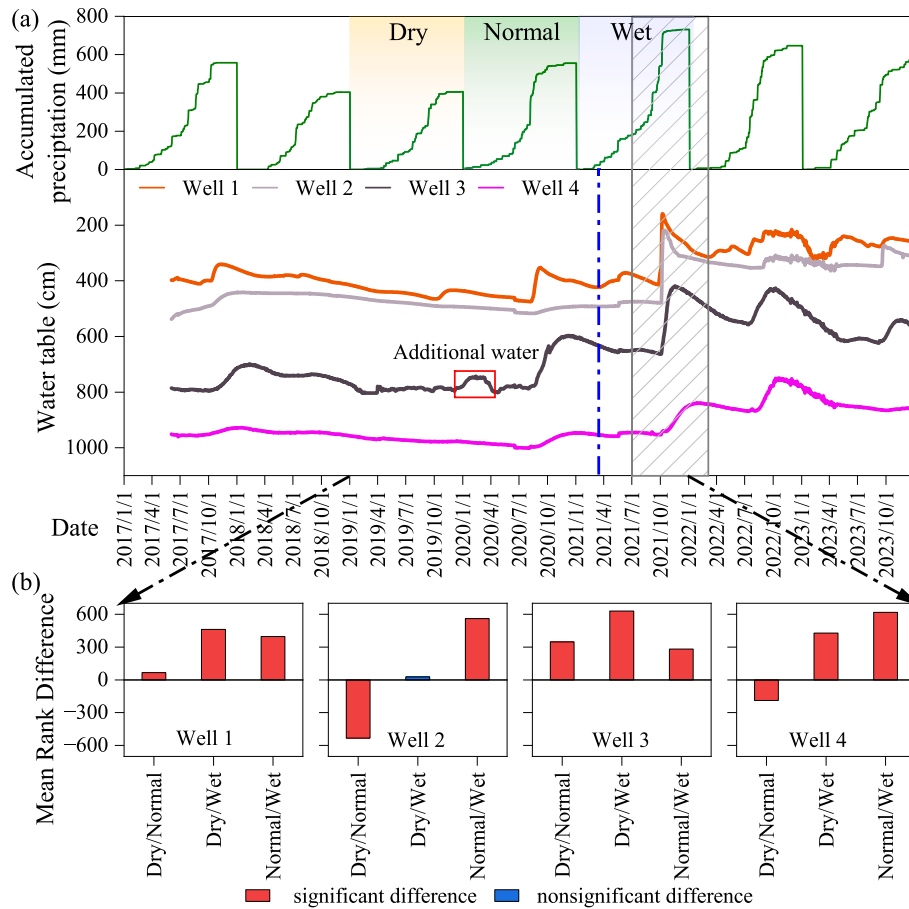


Fig. 8. Temporal dynamic (a) and significant difference in water table among precipitation years (b). Blue dashed-dot line and gray rectangle represent the times of soil water isotopic samplings and monthly measurements at the 4000 cm soil profile, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

average SWC) and P3 (10.2 % average SWC). Notably, P4 induced effective water percolation beyond 500 cm depth along the slope, establishing complete precipitation-soil water-groundwater connectivity within the gully. A similar finding was reported in the previous study, which documented the disappearance of dried soil layers (He et al., 2022). In contrast, P3 only replenished slope soil water depth shallower than the annual average infiltration front, resulting in limited gully recharge. This differential recharge behavior across events corresponds to the saturation zone described in Section 3.2.1, collectively demonstrating event-scale hydraulic connectivity dynamics in this loess region. Furthermore, distinct recharge mechanisms emerged between slope and gully environments, reflecting differential ecological restoration impacts. On slopes, deep-rooted vegetation from the GFG project drove upward soil water redistribution through root stress, consequently decreasing deep percolation (Harmon et al., 2020). Conversely, newly treated land by the GLC project exhibited enhanced hydraulic connectivity through artificial terrain (Zhao et al., 2022). In addition, the observed patchiness of slope soil water suggested that event-scale precipitation predominantly recharges surface soil, while deeper soil and groundwater responses require integrated annual precipitation. These recharge patterns intensified non-linear infiltration of soil water, which is attributed to lateral flow redistribution controlled by topographic and vegetation-mediated modification of soil hydraulic properties through ecological restoration projects (Luo et al., 2023a; He et al., 2024; Liu et al., 2024).

The seasonal variations in SWC across the profile exhibited distinct patterns between non-rainy (November to May) and rainy (June to October) seasons (Fig. 10 and Fig. 11). Two characteristic vertical

profiles emerged, i.e., a unimodal curve peaking at 100 cm depth on the slopes and a generally increased with depth in the gullies. The entire catchment was characterized by similar trend with slope SWC and same significant difference depth with gully SWC, which was attributed to the accounting for area and quantity of SWC. Notably, a buffer layer emerged at 200–300 cm: SWC in the upper profile (0–200 cm) exceeded deeper layers (300–500 cm) during non-rainy season, while this relationship reversed during rainy season. This inversion may arise from enhanced water uptake from deep soil layers (> 200 cm) during non-rainy season, as demonstrated by isotopic tracer studies (Zhao et al., 2020). As a result, SWC frequently fell below the stable field capacity, which promoted the development of a dried soil layer both on the slope and in the catchment. A similar finding was reported in the previous studies, where a common dried soil layer was found to create hydraulic discontinuities that impede vertical exchange between root zones and deeper layers (Bai et al., 2020; He et al., 2022). These results were further confirmed by a significant decrease of deep SWC during non-rainy season ($p < 0.05$) (Fig. 11a). Therefore, two distinct hydrological regimes are identified: slope systems experienced atmospheric coupling limited to the upper 200 cm (average value of SWC ranges from 12.8 % to 15 %), where most water of infiltrated precipitation was recycled via evapotranspiration before reaching deeper layers; gully systems maintained progressive SWC increases with depth, achieving saturation conditions below 300 cm (average value of SWC ranges from 36.6 % to 40.7 %).

In contrast to event-based or seasonal variations, SWC showed no significant differences among precipitation years, maintaining average values ranging from 12.6 % to 32.1 % (Fig. 12). On the slope, the SWC

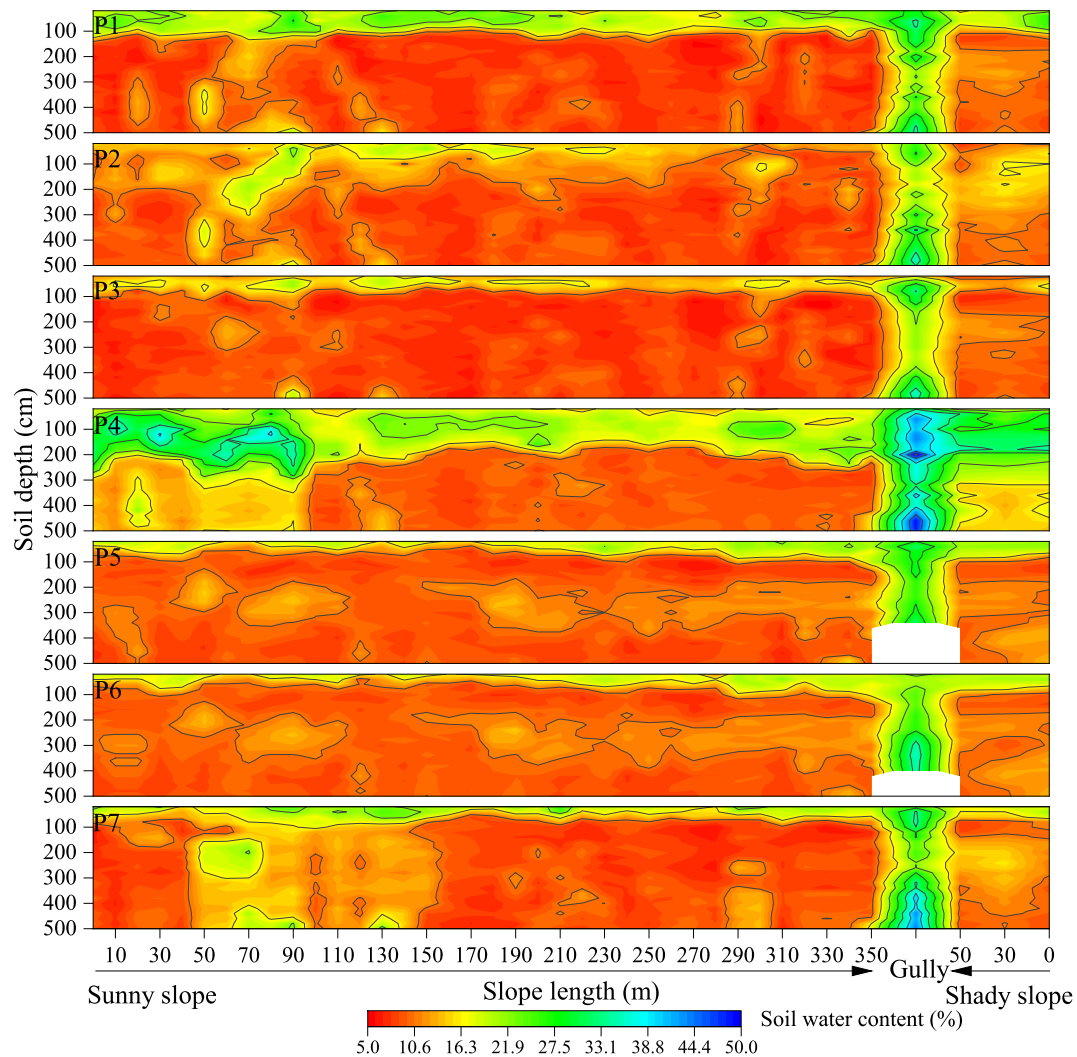


Fig. 9. Soil water content pattern after precipitation event (P) at the 0–500 cm soil profile along the slope length.

was excessively consumed by planted vegetation after the GFG project, leading to remarkably low SWC levels averaging 12.7 %, even during wet years. This persistent imbalance between soil water input and output leads to progressive soil desiccation within the profile, posing a threat to the local ecosystem (He et al., 2022). Notably, gully showed fundamentally distinct hydrological behavior, with average SWC measurements approximately three times higher than those on the slope. Zhao et al. (2019) attributed this spatial heterogeneity to differential water retention capacities enhanced by the GLC project. At the same layer, SWS increased with annual precipitation although the patterns varied across the profile. Specifically, the depths of slope SWS consumed were 200 cm, 300 cm, and 400 cm, corresponding with dry, normal, and wet years, respectively. These findings further suggest that most water from precipitation is stored in the upper soil layer and eventually returned to the atmosphere through evapotranspiration on the slope. Therefore, without considering the contributions from the saturated zone, annual precipitation did not profoundly change SWC, but affected the depth of the hydrological cycle, resulting in negative balances of soil water, which was commonly reported in previous studies (Huang et al., 2024; Wang et al., 2024).

Considering the contribution of the saturated zone, total recharge rates were quantified as 94.6 mm/year (accounting for 23 % of precipitation), 147.8 mm/year (accounting for 27 % of precipitation), and 327.8 mm/year (accounting for 45 % of precipitation) during dry, normal, and wet years, respectively (Table 3), demonstrating the

positive hydrological feedback of ecological restoration projects (Ma et al., 2025). At the annual scale, the total recharge rate was 3.5 times higher in the wet year compared to the dry year, which is consistent with patterns observed in deep-layer (> 200 cm) recharge dynamics (Gao et al., 2023). However, despite positive total recharge values during the dry year, limited deep soil water replenishment (< 200 cm depth) coupled with declining water tables indicated low-efficiency recharge mechanisms. This shallow replenishment depth failed to meet the requirements of deep-root vegetation established through the GFG project (Pei et al., 2025). Meanwhile, accelerated decline in groundwater recharge (60.8 mm/year) highlighted the urgency for improved agriculture water management strategies (Jasechko et al., 2024). Thus, prolonged drought created negative water budgets on the slope and led to a – 15.4 mm/year unsaturated zone recharge rate during the normal year. This is likely because more precipitation is used to replenish soil water rather than moving downward following the GFG project (Chen et al., 2024). Notably, even during the wet year with high precipitation, slope soil water deficits still persist. This paradox may be attributed to enhanced PET (939 mm/year) dominating unsaturated zone water balance in semi-arid regions (Turkeltaub and Bel, 2023). Conversely, the GLC project demonstrated improved hydrological regulation, with unsaturated zone expansion facilitating water storage that supported groundwater recharge rates of 163.2 mm/year and 316.0 mm/year during normal and wet years, respectively. These findings further confirmed the positive effect of the GLC project on soil and water

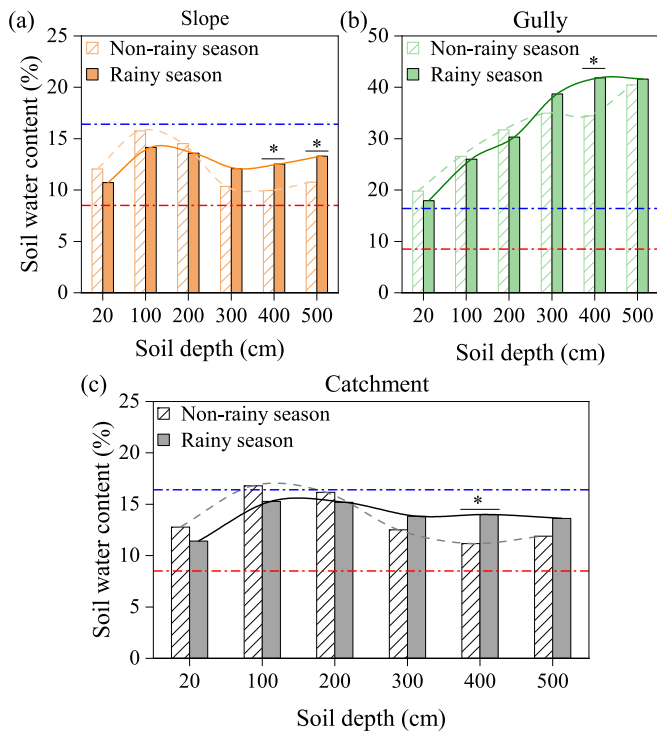


Fig. 10. Soil water content during the non-rainy and rainy seasons at different soil layers. Blue and red dash lines represent stable field capacity and wilting point, respectively. Asterisk indicating significant differences of soil water content between non-rainy and rainy season at $p < 0.05$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

conservation (He et al., 2020). Therefore, it can be inferred that event- and annual-scale precipitation extremes enhanced groundwater recharge processes at the ecological restoration catchment when considering the saturated zone. However, a rise in water table and slope instability can be observed simultaneously after the GLC project, especially during precipitation extremes. These processes can cause soil salinization and soil erosion, threatening crop yield and land development (Jin et al., 2019; He et al., 2020; Cao et al., 2023). Furthermore, a more significant interaction occurred between soil water and chemical components than physical properties (Jin et al., 2025). Consequently, trade-offs should be considered for future ecological restoration

projects. For example, constructing drainage channels on the sunny slopes could help maintain soil and water conservation on the slope while protecting gully soils from salinization.

4. Conclusions

Our study demonstrated that preferential flow pathways and rising water tables create hydraulic connectivity between the unsaturated and saturated zones. Furthermore, precipitation extremes and ecological restoration projects enhance groundwater recharge processes but aggravate recharge heterogeneity. However, these hydrological processes inevitably increase the potential risk of soil salinization, creating a critical trade-off for future ecological restoration projects. Consequently, more hydrological regulation is essential for these systems under both event- and annual-scale precipitation extremes. For instance, constructing drainage channels on sunny slopes (where preferential flow

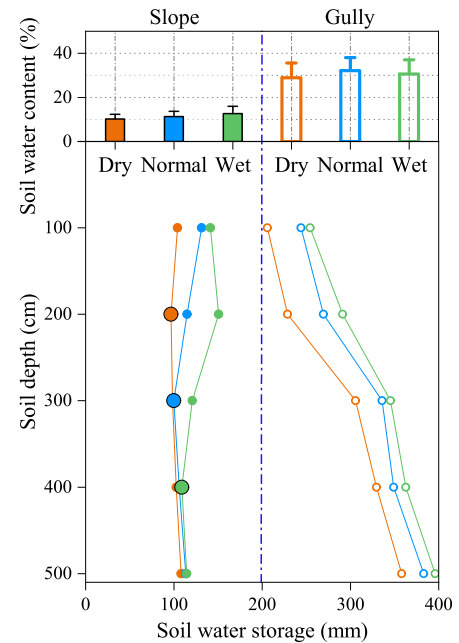


Fig. 12. The soil water content averaged and soil water storage profiles at the slope and gully among dry (orange), normal (blue), and wet (green) years. Error bars represent SD of the mean. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

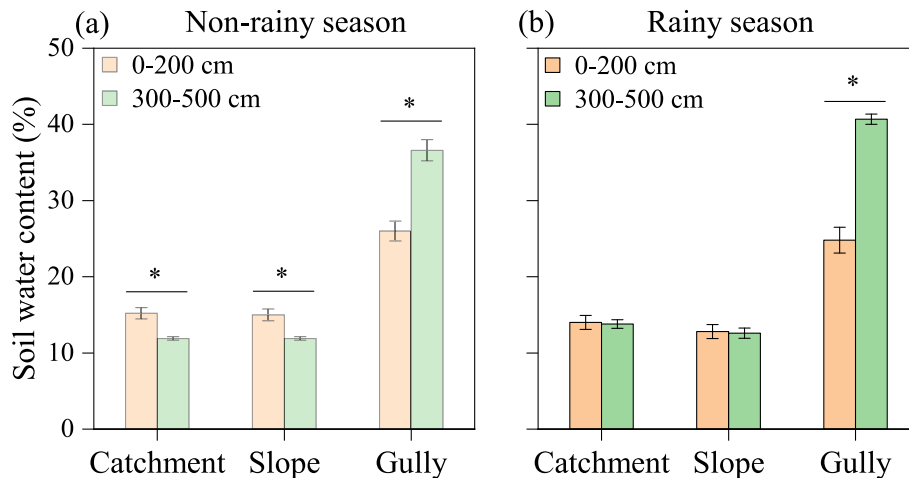


Fig. 11. Soil water content between shallow (0–200 cm) and deep (300–500 cm) during non-rainy (a) and rainy seasons (b). Error bars represent $\pm$ SE of the mean with the asterisk indicating significant differences of soil water content between 0–200 cm and 300–500 cm soil layers at $p < 0.05$.

Table 3
Recharge in unsaturated and saturated zones per unit area within the catchment.

Position	Soil profile (cm)	Recharge rate (mm/year)		
		Dry year (2019)	Normal year (2020)	Wet year (2021)
Unsaturated zone	Slope	0–500	156.3 (6.3)	−22.3 (−0.9)
		Recharged soil layer	0	0
	Gully	0–500	148.0 (5.9)	39.7 (1.6)
		Recharged soil layer	0–180	60–240
Saturated zone	Catchment	0–500	155.4	−15.4
	Gully	Groundwater	−60.8	163.2
Total recharge amount		94.6	147.8	327.8
Account for annual precipitation		23 %	27 %	45 %

Note. Numbers in brackets are average values of soil water storage change.

occurs) could help maintain slope soil and water conservation while protecting gully soils from salinization. These results can provide valuable insights into hydrological processes and contribute to improved groundwater resource management and aquifer protection.

CRedit authorship contribution statement

Meina He: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Yunqiang Wang:** Writing – review & editing, Supervision, Conceptualization. **Yali Zhao:** Writing – review & editing, Resources, Data curation. **Li Wang:** Writing – review & editing, Supervision. **Ziliang Zhang:** Writing – review & editing, Supervision. **Yi Song:** Writing – review & editing, Supervision. **Pingping Zhang:** Writing – review & editing, Supervision, Conceptualization. **Zimin Li:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

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Data availability

Raw data will be available on request.

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